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Progressive collapse fragility of reinforced concrete framed structures through incremental dynamic analysis

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ABSTRACT

In conventional design approaches, structural safety is assessed without considering extreme load conditions that may cause global system collapse as a consequence of local failures in a component or localized portion of the structure. Even when extreme conditions are taken into account, structural safety is not probabilistically assessed and controlled, thus neglecting uncertainties in loads and system capacity. A few studies have been carried out so far, emphasizing the need for probabilistic risk assessment and management of structures to disproportionate (or progressive) collapse. To this aim, fragility analysis may be used to predict the probability of progressive collapse given that local damage has occurred. This approach is a well-established tool in earthquake engineering and explicitly accounts for uncertainties in both demand and capacity, providing the probability of failure as a function of a given intensity measure.

In this paper, fragility functions for low-rise reinforced concrete (RC) framed building structures are presented to be implemented in progressive collapse risk assessment. Two building classes representative of European buildings designed for gravity loads and earthquake resistance in accordance with Eurocodes 2 and 8, respectively, were investigated. Fiber-based finite element (FE) models were developed and integrated with numerical techniques able to simulate the removal of first-story columns within an open source platform. Nonlinear response, resisting mechanisms and damage patterns under sudden column loss scenarios were reproduced at both local and global structural levels. Based on statistics and probability distribution functions for geometry, material properties and loads of the case-study building classes, Monte Carlo simulation was performed to generate random realizations of structural models. Fragility functions at multiple damage states show a significant influence of both seismic design/detailing and secondary beams on robustness of the case-study RC building classes.

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1. Introduction

In the last decades, iconic and public buildings in urban environment have been subjected to accidental and man-made extreme events such as fires, impacts or explosions, thus triggering increased interest in blast- and progressive collapse-resistant building analysis and design [1–4]. Several buildings around the world have experienced partial or total progressive collapse as a result of deliberate terrorist attacks, uncontrolled gas releases, and vehicle or aircraft impacts, confirming that structural systems designed according to conventional codes are not necessarily able to withstand abnormal loading conditions. Although building collapse is a rare event, it may result in significant casualties and property loss when it occurs [5] and such low probability/high consequence events can involve both industrial and residential structures. In light of this scenario, homeland security have become a primary concern for public authorities and stakeholders, seeking for passive and active strengthening strategies of different structural typologies ranging from civil to strategic and military facilities.

Several definitions of disproportionate (or progressive) collapse have appeared in the literature [5–11] and the role of non-structural protective measures such as barriers, sacrificial elements and control of public access has been recognized to be crucial for a cost-effective mitigation of progressive collapse risk. On the other

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hand, simple and conservative design methodologies to ensure resistance against progressive collapse phenomena have been proposed in modern codes and standards [12–15], focusing on the performance of structural components. Robustness, integrity, continuity, redundancy and ductility have been reaffirmed to be key structural requirements for an adequate response under column loss conditions [16–18]. While direct and indirect design methods to enhance resistance to progressive collapse are discussed in guideline documents [13,14], a large number of research contributions have examined and proposed modeling and numerical techniques for progressive collapse analysis of reinforced concrete (RC), steel and composite framed structures and structural subassemblies [10,11,19–32]. Procedures for FE [24,27,32], lumped-plasticity [17,21–23,25,26] and spread-plasticity [10,11,16,18,24,28–31] progressive collapse analysis have been explored through general purpose codes or open platforms, thus quantifying the influence of numerical approaches with different levels of sophistication. In particular Li et al. [31] validated the application of fiber-based models for progressive collapse analysis, comparing experimental tests and numerical simulations. Using a fiber-based approach, Brunesi and Nascimbene [18] compared different FE codes in order to additionally investigate different analysis types and algorithms, as well as element formulations and removal techniques.

Besides extensive efforts in design and simulation, structural safety for extreme loads that may lead to progressive collapse has been probabilistically assessed and controlled in a few cases, thus neglecting uncertainties in loads and system capacity. To this aim, this study is focused on the development of fragility models for low-rise RC framed bare buildings which may be applied for disproportionate collapse risk assessment and management. First of all, the progressive collapse sensitivity to the nonlinear analysis method and structural modeling technique was assessed in the case of two structures with deterministic features. The former structure was designed only for gravity loads in compliance with Eurocode 2 (EC2) [33], and the latter was designed for earthquake resistance according to Eurocode 8 (EC8) [34]. Then, the analysis moved from a deterministic to a probabilistic fashion to assess the progressive collapse fragility of the case-study structures. The following 4-story building classes were investigated: EC2-conforming RC buildings and EC8-conforming RC buildings. For each building class, fragility analysis was based on (i) the random generation of two-dimensional (2D) and three-dimensional (3D) fiber-based models through Monte Carlo simulation, (ii) incremental-mass nonlinear time-history analysis (NLTHA) of each structural model, and (iii) derivation of fragility functions for each structural model and damage state of interest. Finally, the sensitivity of progressive collapse fragility to design rules (i.e. seismic or non-seismic) and redistribution capacity of secondary beam systems was quantified.

2. Methodology

Fragility models are a well-established tool in earthquake engineering, providing estimates of the proportion of a building class falling within discrete damage bands from a specified earthquake demand [35–40]. Those curves relate the probability of exceedance of multiple damage states to a ground motion intensity measure (IM) such as peak ground acceleration (PGA), so they are a fundamental component to describe the physical vulnerability of a population of buildings. If seismic vulnerability is a measure of how prone a building is to damage for a given severity of ground shaking, fragility curves may be regarded as a graphical representation of seismic vulnerability [33], explicitly accounting for uncertainties in demand and capacity. As discussed in [37–39], several approaches have been proposed in the past to provide a mathematical formulation to vulnerability, either in terms of damage probability matrices or vulnerability curves. Furthermore, there is not a general consensus on a unique methodology for derivation of vulnerability models to be used in regional earthquake loss assessment, due to a variety of structural modeling options, analysis techniques, and damage criteria. Although any model with its level of sophistication has limitations and advantages, the development of fragility models is recognized to be a crucial aspect for disaster mitigation actions. By contrast, a few research works have appeared in the literature so far (see e.g. [41–43]), emphasizing the need for probabilistic risk assessment and management of disproportionate collapse under blast loads and/or sudden column loss scenarios.

A framework for progressive collapse fragility analysis of RC framed structures is presented herein (Fig. 1), using concepts of methodologies developed for earthquake engineering applications and specific simulation techniques for progressive collapse analysis (Fig. 2). The flowchart presented in Fig. 1 outlines the components included in the probabilistic mechanics-based assessment procedure used for the analytical derivation of fragility curves able to characterize the conditional probability of exceeding a certain damage level for a prescribed load intensity. The conditional probability may be defined as follows:

$$P_{ik} = P[D \geqslant d_i | S = s_k] \tag{1}$$

where D is the damage measure (DM) used to describe structural or nonstructural damage, S is the scalar or vector-valued IM used to describe the load intensity, and P_{ik} is the probability that the damage level d_i is reached or exceeded for a given IM level s_k.

After that computational models and nonlinear analysis methods were identified as discussed below, the generation of the building population implied modeling of random variables (RVs) in terms of material properties, geometrical parameters, and design loads. Based on statistics and probability distributions for each RV, several thousands of structural models for each building class was randomly generated via Monte Carlo simulation. Simulated design, numerical modeling and damage analysis of any structural model were then carried out to assess the potential for progressive collapse. In this work, two design rules (i.e. according to EC2 [33] and EC8 [34]) and two structural idealizations (i.e. 2D and 3D models) were considered, resulting in four types of structural models each of them characterized by different kinds of uncertainty. While the definition of both column loss scenario and reference IM is required to perform nonlinear analysis, the selection of criteria for the identification of damage states is a crucial aspect for an accurate and reliable evaluation of structural performance. After that capacity and demand were convolved, the distribution of buildings in each damage state was used to derive the statistical parameters – i.e. mean (μ), standard deviation (σ) and coefficient of determination (R^2) – of each lognormal fragility function, fitting cumulative fragility points by means of nonlinear regression analysis.

In accordance with the classification summarized in [19], four analysis types can be used in both direct and indirect progressive collapse-resistant design/assessment procedures: linear static analysis, linear dynamic analysis, nonlinear static (or pushdown) analysis, and nonlinear dynamic analysis. The disadvantage of a linear simulation, either static or dynamic, is the inability to include material and geometric nonlinearities such as large displacements/rotations (i.e. beam-catenary action), second order effects, coupled axial-flexural inelastic behavior and plastic hinge formation (i.e. stiffness/strength degradation and ductility exploitation). Nonlinear static analysis is relatively simple and provides a load–displacement capacity curve that, similarly to seismic analysis, allows one to assess whether a structure has

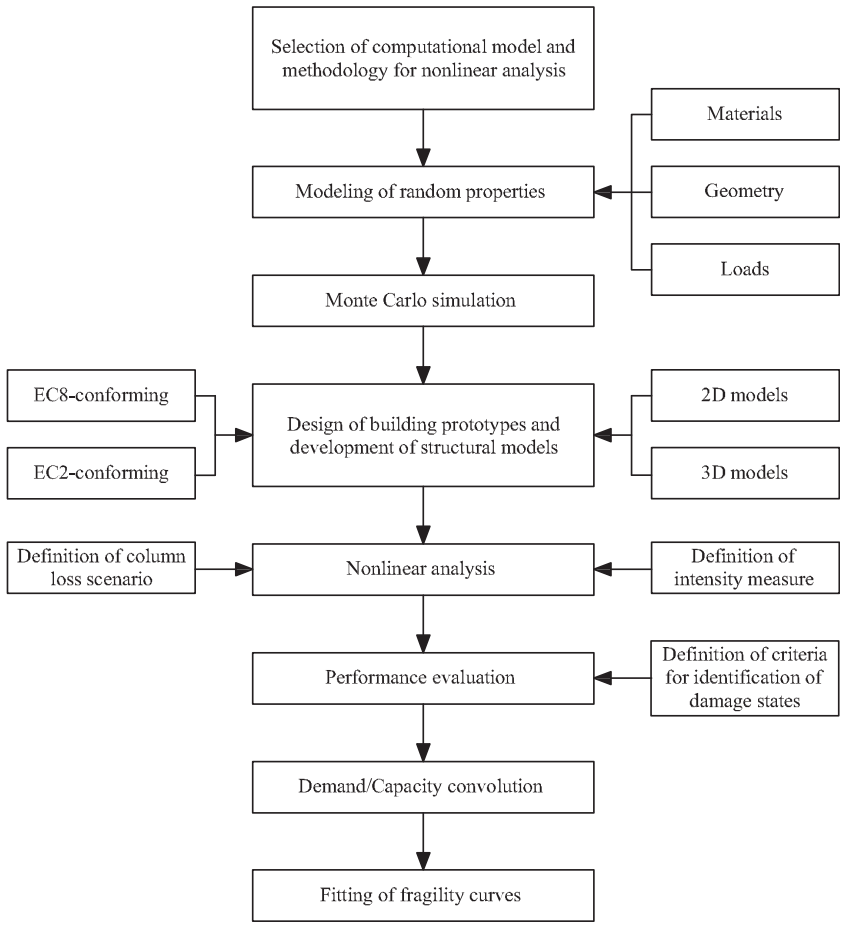

Fig. 1. Flowchart of progressive collapse fragility analysis.

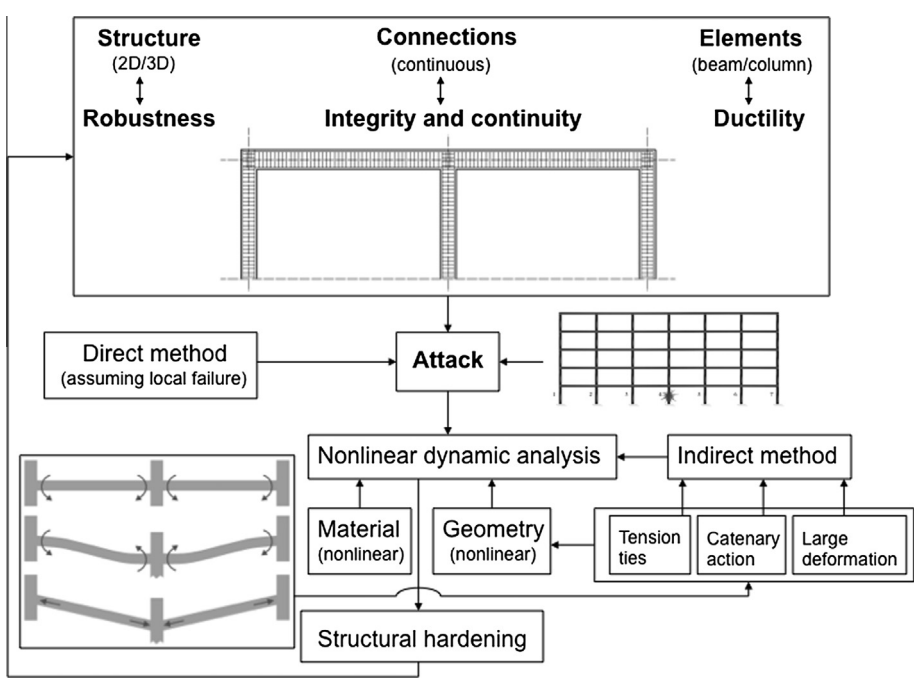

Fig. 2. Framework for progressive collapse resistance simulation/assessment [18].

adequate capacity to resist extreme loads or not, in a static fashion. Nonetheless, one determining factor in considering that a local portion of the structure has failed is the highly dynamic effect produced when a structural element is rapidly removed from the

frame. As discussed in [44], there are structures which are statically safe but dynamically unsafe due to the fact that the time-dependent overloads caused by element removal may induce the progressive fracture of other members before a new equilibrium state is achieved (i.e. cascade or domino effects). This requires the nonlinear dynamic behavior of a structure to be taken into account in progressive collapse simulations [18]. In addition, comparative studies between nonlinear static and dynamic simulations have been performed, showing the former approach to be highly non-conservative for progressive collapse assessment at low-medium displacement levels where immediate occupancy and serviceability limit states govern structural performance [18,25]. To characterize the nonlinear response of the random populations of case-study buildings, dynamic analysis was carried out for each structure realization. As fragility analysis needs an iterative simulation technique in this case, a series of incremental-mass NLTHAs were performed by factorization of the code-compliant load combination assumed for progressive collapse applications [13]. To that aim, a step function with monotonically increasing magnitude till failure was used to simulate different dynamic loading conditions, as done in similar earthquake engineering applications by scaling the input ground motion [45–49].

After a brief description of FE representation, column-removal strategy and methodologies adopted to design and detail the case-study buildings, some numerical aspects are further discussed in the following section assuming two reference structures to compare different simulation procedures. Those structures had the same overall geometry but different detailing as they were separately designed in compliance with EC2 [33] and EC8 [34].

3. Comparison of different computational strategies

The buildings under study were 4-story, 4 × 4-bay RC framed structures. Both 2D and 3D fiber-based models were developed for each case-study building, that is EC2-conforming or EC8-conforming building. Then, each structural model was separately assessed through pushdown and incremental dynamic analyses (IDA). This allowed the authors to carry out a two-way comparison between response predictions obtained (i) by different design solutions for the same type of analysis (i.e. EC2-conforming versus EC8-conforming), and (ii) by different analysis types for the same design rules, in direct relation to the type of structural model under consideration (i.e. 2D versus 3D). The conclusions drawn from capacity curves presented in this section were then extended to the case-study building classes in terms of fragility models and are discussed in Section 4.

3.1. Structural models

Fig. 3 shows isometric and plan views of the low-rise buildings under study, which had a lateral load-resisting system composed of five primary frames connected each other by one-way RC joist slabs and continuous cast-in-situ secondary beams. The center-to-center plan dimensions were 24 m in length (x-direction) and 16 m in width (y-direction) at any floor, with column spacing of 6 m and 4 m in longitudinal (L_x) and transverse (L_y) directions, respectively. The interstory height was 3 m at each floor. In both EC8-conforming and EC2-conforming structures, columns and beams had $400 \times 400\ \text{mm}^2$ and $500 \times 300\ \text{mm}^2$ cross sections, respectively. Assuming buildings to be employed for housing, dead (DL) and live (LL) loads were supposed to be 3 kN/m^2 and 2 kN/m^2, thus implying a downward load on beams (Q_b) for progressive collapse simulation of 14 kN/m in both static and dynamic fashion, as expressed in Eq. (2):

$$Q_b = \text{DL} + 0.25\text{LL} = 14\ \text{kN/m} \tag{2}$$

Even if a properly detailed slab may potentially improve structural response contributing to the redistribution mechanism of the entire building, a one-way RC floor slab was conservatively assumed to provide no resistance against progressive collapse while its weight and inertia were implicitly included in FE simulations. This approach was also used to account for partition walls. Material properties were set to 25 MPa for the characteristic compressive cylinder strength of concrete (f_{ck}) and 450 MPa for the characteristic yield strength of both longitudinal and transverse reinforcements (f_{yk}). The simulated design of the EC2-conforming and EC8-conforming structures was carried out using design strengths according to EC2 [33], whereas mean strengths were selected for nonlinear analysis. In particular, the design strengths of concrete (f_{cd}) and reinforcing steel (f_{yd}) were obtained as $f_{cd} = 0.85\ f_{ck}/1.5$ and $f_{yd} = f_{yk}/1.15$, respectively. Mean compressive strength of concrete (f_{cm}) and mean yield strength of reinforcing steel (f_{ym}) were assumed to be $f_{cm} = f_{ck} + 8$ and $f_{ym} = f_{yk} * 1.1$, respectively.

The EC8-conforming structure was designed for medium-high seismicity assuming the PGA at bedrock to be equal to 0.30g for life safety limit state. A type C ground (180 m/s < V_s < 360 m/s) was selected to perform design in medium ductility class by a series of response spectrum analyses on 3D models, incorporating accidental eccentricity of the center of mass. An equivalent viscous damping of 5% was chosen to construct type 1 design spectra in accordance with current European prescriptions [34]. As a result of those assumptions, a uniform longitudinal reinforcement consisting of 12Ø20 bars was provided in columns (where Ø stands for bar diameter), while 6Ø20 + 4Ø10 were used as longitudinal reinforcement of beams, thus implying longitudinal reinforcement ratios of approximately 1.5% and 2.4% for beams and columns, respectively. The transverse reinforcement was composed of Ø10 stirrups with spacing of 50 mm and 80 mm at both end zones of columns and beams, respectively. Rebar cover was everywhere set to 40 mm.

In addition to the aforementioned dead and live loads, snow and wind loads equal to 1.6 kN/m^2 and 0.9 kN/m^2 were assumed to design the EC2-conforming structure. All beams and columns were assumed to have Ø16 longitudinal bars and Ø8 stirrups with 100 mm spacing at both end zones. In the case of columns, 8Ø16 longitudinal bars were uniformly arranged along their length. In the case of beams, longitudinal reinforcement detailing was different at end zones and in the proximity of mid-span. In fact, 5Ø16 and 3Ø16 bars were respectively provided at the top and bottom of cross sections at both end zones, whereas 2Ø16 bars and 3Ø16 were respectively provided at the top and bottom of cross sections at mid-span. Finally, 4Ø10 longitudinal bars were uniformly distributed along the beam length in order to satisfy minimum spacing requirements between longitudinal bars. Longitudinal reinforcement ratios of about 1.3% and 1.0% were then respectively provided in beams and columns, thus resulting in a 13% and 57% decrease with respect to the EC8-conforming structure.

Table 1 outlines the section properties of beams and columns in terms of member size and reinforcement layout, while Fig. 4 sketches complementary design drawings of both building prototypes. The main assumptions concerning the assessment strategy, modeling approach and numerical column-removal techniques are discussed in the following.

3.2. Numerical techniques

Numerical idealizations developed along the lines of classical fiber force-based approaches [50], commonly used in seismic analysis of steel [47,51] and RC [36,38,39] structures, have been prepared within the open FE code SeismoStruct [52] in order to predict the nonlinear response of the two case-study buildings,

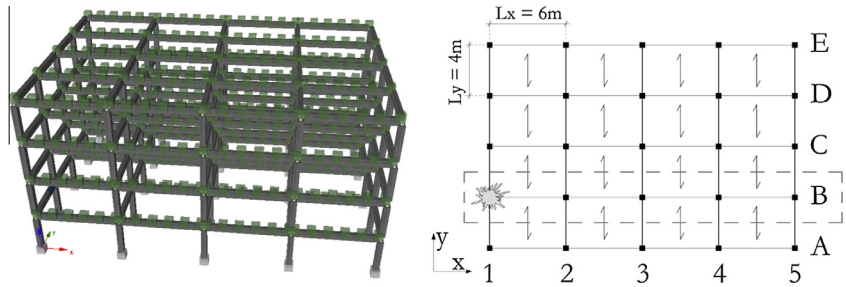

Fig. 3. Isometric and plan views of case-study building model.

Table 1
RC section properties of EC8-conforming and EC2-conforming building models.

Building class	Element	Plan and floor location	Size (mm^2)	Longitudinal reinforcement	Transverse reinforcement
EC8-conforming	Beam	Any line/any floor	500 × 300	3Ø20 + 4Ø10 + 3Ø20	2-leg Ø10@80
	Column	Any line/any floor	400 × 400	12Ø20	2-leg Ø10@50
EC2-conforming	Beam	Any line/any floor	500 × 300	5Ø16 + 4Ø10 + 3Ø16	2-leg Ø8@100
	Column	Any line/any floor	400 × 400	8Ø16	2-leg Ø8@100

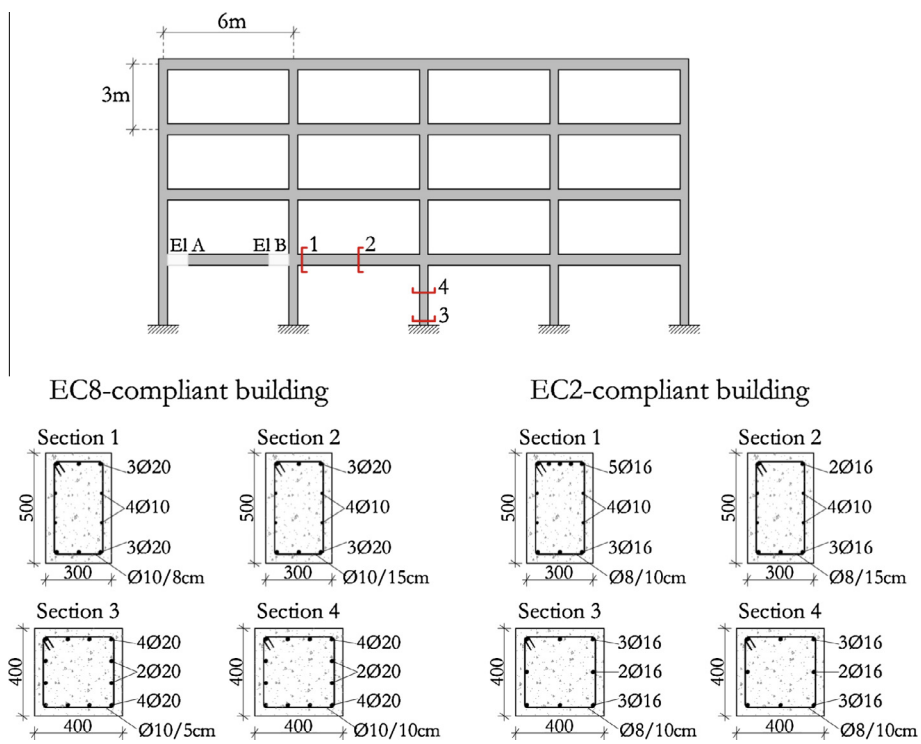

Fig. 4. Reference building prototype: cross sections and reinforcement arrangement of EC8-compliant structure and EC2-compliant structure.

in static and dynamic fashion. Inelastic 3D beam-column fiber elements were assumed to model the frame members, explicitly including geometric and material nonlinearities. In detail, a total corotational transformation was used to account for potential large displacements/rotations and P-Delta effects expected during progressive collapse analysis. The implementation of total corotational transformation is based on the exact description of kinematic transformations associated with large displacements and 3D rotations of the frame member. A distributed inelasticity approach, where sectional stresses and strains are obtained through direct integration of the uniaxial material response of individual fibers, was adopted to simulate the spreading of inelasticity over the member length and cross section.

Even though a force-based formulation does not necessarily require element discretization [50], a one-to-six correspondence between structural members and model elements (each of them having 5 integration points and 400 fibers) was assumed to accurately accommodate the deformed shape during progressive collapse analysis. A simple bilinear constitutive model with isotropic strain hardening was assigned to reinforcing steel, whereas the uniaxial uniform confinement model by Mander et al. [53] was used to simulate the cyclic behavior of concrete, explicitly accounting for tension softening and conservatively omitting strain rate effects in the simulation [54]. NLTHA was carried out using a tangent stiffness-proportional Rayleigh damping and the Newton–Raphson algorithm with a displacement/rotation-based

convergence criterion to iteratively equilibrate loads. As discussed in [18], a special purpose death element routine (i.e. activation/deactivation time option) was integrated in SeismoStruct [52] to operate member removal, thus enabling the authors to dynamically simulate the overall structural response and damage patterns/mechanisms under a threat-independent sudden column loss scenario. The ultimate capacity of the two case-study buildings was defined in terms of steel and concrete strains. The fracture/buckling strain of reinforcing steel was conservatively set to 6%. The ultimate compressive strain of concrete was obtained in accordance with Mander et al. [53], resulting in 0.6% and 0.8% for the EC2-conforming and EC8-conforming case-study structures, respectively, because of the different confinement effects provided by stirrups. Therefore, the first exceedance of those strain limits in the first cross section of the frame was conventionally interpreted as a conservative check of the "near collapse" limit state. Similar approach and performance criteria were assumed in past studies [35,36,38,39] to identify extensive/complete damage state for similar RC framed structures. In addition, code-compliant shear capacity verifications were included in the simulations to notify whether demand exceeded capacity; in that case, shear strength was reduced by 50% [17,33,34,55]. Nonetheless, conventional strength hierarchies for capacity design of the EC8-conforming structure [34] prevented any brittle and non-dissipative shear failure. A similar consideration can be drawn for the EC2-conforming structure, whose collapse mechanism was still governed by a combination of flexural–axial hinging as detailed later on. Offline computations and safety verifications were performed to ensure any potential local mechanism not to occur in beam-column joints [17,33,34], which were modeled as rigid elements.

To assess and compare the progressive collapse potentials of the two case-study structures in terms of analysis method and structural modeling, an internal 2D frame was first extracted from each 3D model and then analyzed by removing the leftmost ground column. That scenario was considered as it is the most demanding single column-removal condition for a framed structure [18]. Response predictions for the 2D model were then collected and compared with those provided by nonlinear analysis of 3D models, in order to quantify the effects of secondary beam systems in the redistribution of gravity loads from a removed column.

3.3. Results and discussion

An example of the deformed shapes predicted at ultimate conditions by nonlinear dynamic analysis is provided in Fig. 5, where the failure mode obtained for the EC8-conforming and EC2-conforming structures is additionally shown and compared in terms of flexural hinging mechanism and arch effect/catenary action developed in the critical portions of the beam at the first floor (i.e. key elements labeled as "El. A" and "El. B"). In particular, Fig. 5(a) and (b) reveals the frames to experience almost identical ultimate vertical displacements at the location of the removed column, but different resisting mechanisms are forced to occur in beams and columns as a consequence of their different detailing. This trend is confirmed and corroborated even further by Fig. 5 (c) and (d), presenting moment–rotation and axial force–vertical displacement curves in key elements of both frame models. As the longitudinal reinforcement arrangement of the EC8-conforming structure was symmetrical, the change in bending moment sign caused by column removal produced a demand reduction and a safe shift in the moment–rotation curve, as discussed later on in more details. That outcome is consistent with the assumption of RC cross sections with identical capacity under loading reversals. As shown in Fig. 5(c), the moment–rotation curve reaches its unloading branch and oscillates upward and downward when the structure oscillates around its displacement peak. Therefore, the reinforcement layout based on seismic design criteria and good-design practice rules forced the primary frame to fail by concrete crushing at the beam end located on the opposite side with respect to the removed column (i.e. "El. B" rather than "El. A"). In detail, a peak chord rotation of up to 0.022 rad, namely close to rotational capacity of the cross section, was observed in "El. B" whereas a more than halved demand (i.e. 0.010 rad) was found in "El. A". By contrast, when the EC2-conforming frame was considered, a similar chord rotation of about 0.012 rad was found in "El. A" and "El. B", thus resulting in a much higher discrepancy in bending moment demand between these two elements, in comparison with demand on the EC8-conforming frame.

As depicted in Fig. 5(d), axial compressive forces took place after column removal in the beams of the first floor, due to the arch resisting mechanism that developed between internal and external joints. Stresses increased until concrete crushing and cracking occurred in the compressive and tensile regions, respectively. Therefore, a plastic hinge formed and limited the sectional resisting capacity. Cracking propagated further through the beam section depth and catenary action decreased the compressive forces. While moderate tension appeared in the beams of the second floor, that observed in first-floor beams was found to fluctuate being they too damaged and hence unable to react. That mechanism was more pronounced in the EC8-conforming frame, showing axial force peaks that were roughly 60% higher than those predicted in the case of the EC2-conforming frame.

To further detail the progressive collapse mechanism observed in case of seismic design, Fig. 6 shows its time-dependent evolution in terms of bending moment distribution in the beams of the first floor and moment–rotation curves in their critical portions. In particular, the moment demand after column removal is collected at peak displacement, at the end of the analysis and in the case of two intermediate time-instants corresponding to pre- and post-peak displacements. A classical bending moment distribution, with positive peaks at beam ends and negative ones at midspan, is observed in static conditions. After column removal, the structure starts falling to achieve its peak vertical displacement and oscillate around it, thus implying a change in bending moment sign (i.e. from positive to negative) at "El. A", while the bending moment demand recorded in "El. B" keeps on increasing with the same sign (i.e. positive). In addition to that, "El. A" and "El. B" have the same potential for progressive collapse resistance, being their RC cross sections characterized by an identical capacity under loading reversals, because of the symmetrical longitudinal reinforcement arrangement assumed in case of the EC8-compliant structure. This prevents failure from occurring in "El. A" and permits one identifying the beam end opposite to the column removed as the critical portion of the structure when simple seismic detailing is provided.

The vertical displacement time-histories obtained by IDA on 2D models are plotted in Fig. 7 to compare the global responses of the two reference frames, confirming the considerations previously drawn in terms of local resisting mechanisms. Even if both framed systems failed by concrete crushing in the same zone for similar vertical displacements, their collapse modes were characterized by different levels of resistance, robustness, rationale and control. The maximum dynamic loading capacity was found to be $0.55Q_b$ and $0.42Q_b$ for the EC8-conforming and EC2-conforming frame, respectively. As the scale factor (SF) of Q_b was increased, vertical displacement increased and damage propagated in the structure, as confirmed by a significant period elongation shown in the set of time-history curves and particularly evident for the EC2-conforming structure. A comparison between the displacement histories of the two frames at the same dynamic load level reveals the response of the EC2-conforming structure to be significantly

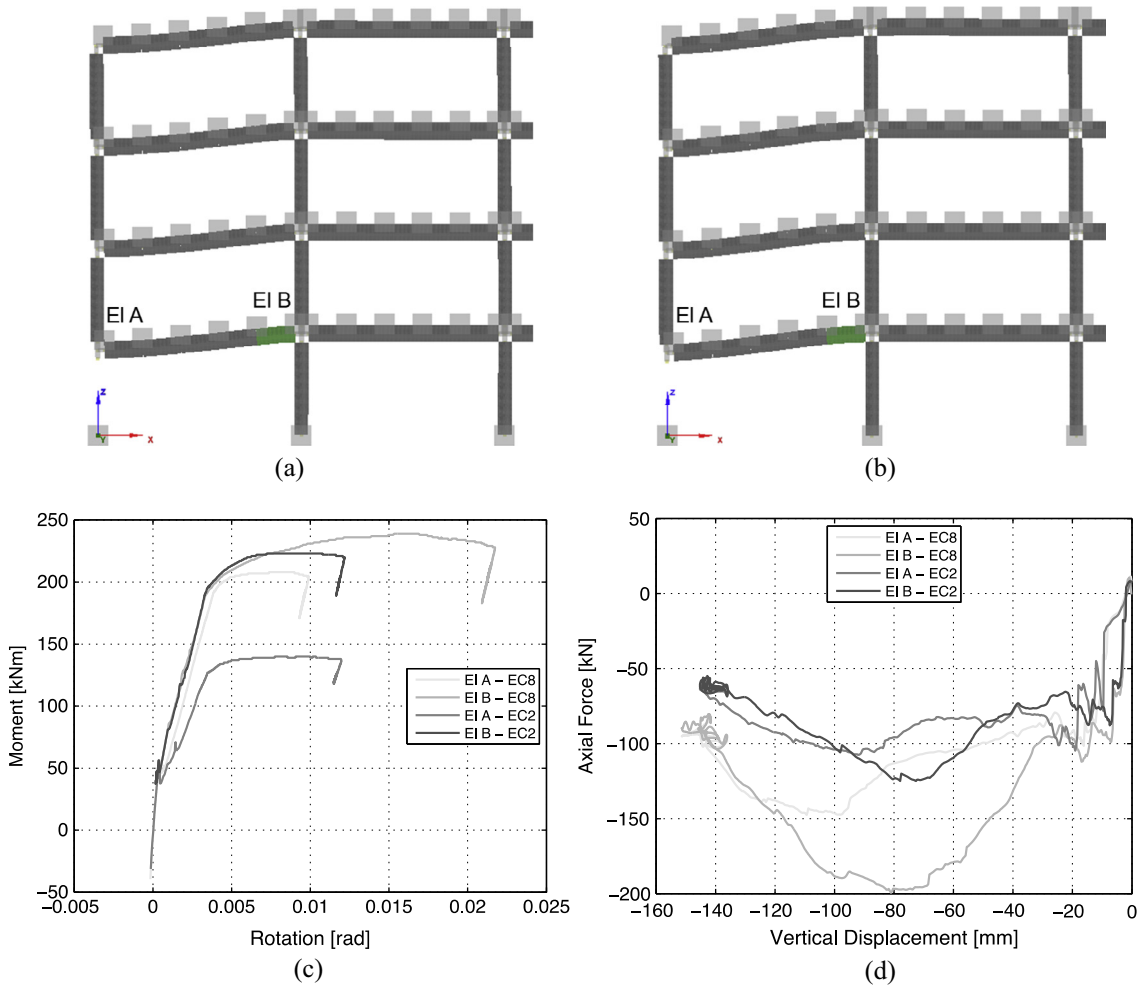

Fig. 5. NLTHA deformed shapes at ultimate conditions of (a) EC8-conforming building and (b) EC2-conforming building; (c) moment–rotation and (d) axial force–vertical displacement curves of key elements. Note: the vertical displacement shown in (d) are recorded in correspondence to the column removed.

more weakened in character. For a given load of $0.42Q_b$, the EC2-compliant building is severely damaged and close to collapse, while the EC8-conforming structure is still able to displace almost elastically and move further in the nonlinear regime of its response. At this stage (SF = 0.42) peaks of up to 93.3 mm and 145.3 mm were predicted for the EC8- and EC2-compliant buildings, thus showing the latter to experience displacements roughly 36% higher than the former because of the benefits from seismic design/detailing. Alternatively, the EC8-conforming structure can carry higher loads for a given vertical displacement, as shown when discussing the resisting mechanism of the two prototypes at their ultimate conditions (see Fig. 5). In this case the EC8-compliant building revealed displacements only 5% higher than those observed in the EC2-conforming frame, as a consequence of slightly different ultimate concrete strains assumed according to the different confinement effects provided by the stirrups of their cross sections, but presented a more rationally controlled collapse mode that was characterized by a much higher potential for progressive collapse resistance.

Furthermore, the displacement peak of each dynamic simulation was stored to construct the load–displacement capacity curves provided in Fig. 8, where IDA envelopes are compared to results of a displacement-controlled pushdown analysis based on the application of an incremental uniform load over the beams [17], because a further aim of this comparative study is to estimate the dynamic amplification factor of the two prototypes. As previously mentioned, nonlinear static analysis visibly overestimates the ultimate capacity of both structures, leading to maxima of up to $0.77Q_b$ and $0.92Q_b$ for the EC2-conforming and EC8-conforming frames, respectively. In particular, peak resistance predictions by pushdown analysis were found to be 83% and 67% higher than those provided by NLTHA. Pushdown analysis predictions were even less conservative at lower displacement levels, that is when structural performance was characterized by minor or moderate damage. The main difference between the two simulation techniques is in the fact that pushdown analysis accounts for dynamic effects neither in a dynamic fashion [44] nor in a simplified manner [14]. A different dissipative mechanism and related propagation of damage through the cross section depth are also forced to occur by different loading procedures, since pushdown analysis is based on the application of a monotonically increased downward load or load distribution, while during dynamic analysis the structure is damaged under cyclic reversals, oscillating upward and downward in such a way that more closely reproduces the physical nature of progressive collapse. Regardless of the analysis method, the ultimate capacity of the EC8-conforming frame was approximately 20–30% higher than in case of EC2-conforming frame, reflecting a benefit from seismic design rules.

After that robustness of 2D frames taken out from the EC2-conforming and EC8-conforming structures was assessed, the focus of the research moved to the influence of secondary framing beams on progressive collapse potential. Those beams

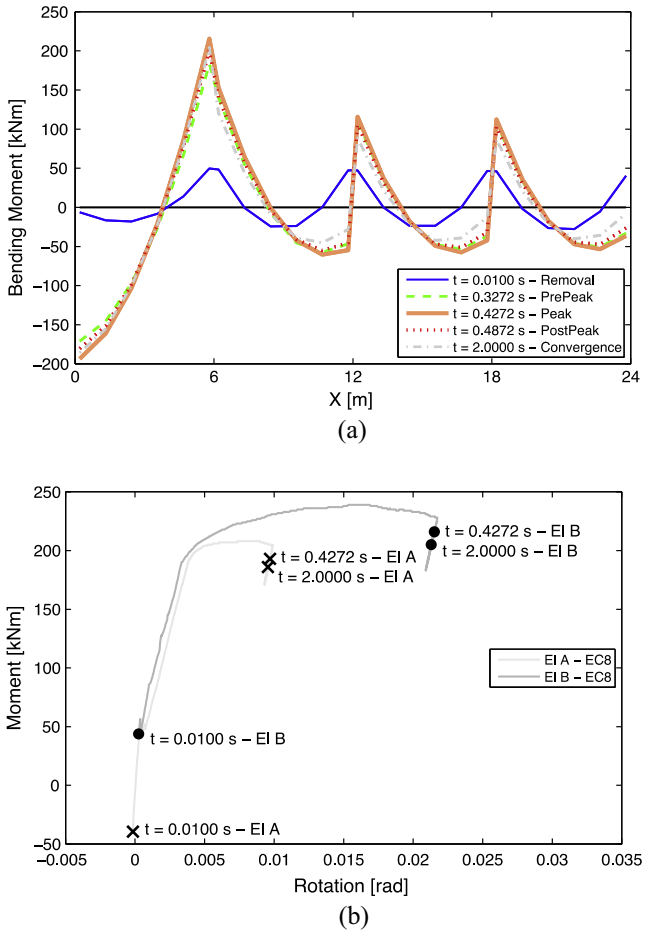

Fig. 6. Time-dependent evolution of progressive collapse mechanism: (a) bending moment demands in the beams of the first floor and (b) moment–rotation curves in their critical portions. Note: EC8-compliant 2D frame.

may possibly be considered as a latent resource of stiffness and strength in order to provide a rationally-controlled alternative load path for the unbalanced demand induced by sudden column loss scenarios. Given that one-way slabs were assumed according to construction practice for such buildings, secondary beams were supposed to resist only their self-weight so they were designed to comply with minimum reinforcement requirements by current European standards [33,34]. This implied symmetrical longitudinal reinforcement ratio of 1% to meet minimum seismic provisions and a slightly less than halved ratio in case of the EC2-conforming design. The cross section of those members was kept constant and equal to that assumed for the primary beam system of both structures, as specified in Table 1 and Fig. 4.

The redundancy provided by secondary beams or "secondary-beam effect" significantly improved the robustness of both case-study structures, being their contribution effective in controlling the overall structural response and providing an alternative load path for column loss-induced overloads (see Figs. 9 and 10). As shown in Fig. 9, displacement time-history curves smoother than those determined for 2D models were predicted for both building models because these secondary members may be actively used to interact with the primary frames, finally resulting in a stiffer, stronger and more stable resisting/dissipative mechanism. Due to their visible participation in primary mechanism, vertical displacement peaks of up to 18 mm and 14 mm were obtained for the EC8-conforming and EC2-conforming structures, respectively, highlighting maxima that are approximately one order of magnitude lower than those obtained on 2D models (see Figs. 7 and 9). In addition, the IDA curves presented in Fig. 10 reveal an ultimate progressive collapse resistance equal to $1.35Q_b$ in case of EC8-conforming structure and $0.95Q_b$ in case of EC2-conforming structure, evidencing that the robustness of the earthquake-resistant structure was about 42% higher. Such an outcome confirms the considerable contribution provided by secondary frame elements for gravity load redistribution as a consequence of a sudden column-removal condition. It is also found from Figs. 10 and 8 that:

$$\left(\frac{Q_{b,St}}{Q_{b,Dyn}}\right)_{3D} < \left(\frac{Q_{b,St}}{Q_{b,Dyn}}\right)_{2D} \tag{3}$$

where $Q_{b,St}$ and $Q_{b,Dyn}$ are the static and dynamic capacities, while $(Q_{b,St}/Q_{b,Dyn})_{3D}$ and $(Q_{b,St}/Q_{b,Dyn})_{2D}$ are the static-to-dynamic capacity ratios of 3D and 2D models, respectively.

In other words, pushdown analysis was slightly more conservative than in case of 2D models, highlighting overestimates (at ultimate conditions) of up to 35% and 59% for EC8-conforming and EC2-conforming structures, respectively. Fig. 11 shows the force-based dynamic amplification factor (DAF) versus the dynamic vertical displacement of the structure, be it designed for gravity loads or earthquake resistance. DAF was computed as the ratio between static and dynamic capacities in terms of vertical peak resistance of the 2D or 3D structural model. DAF is a

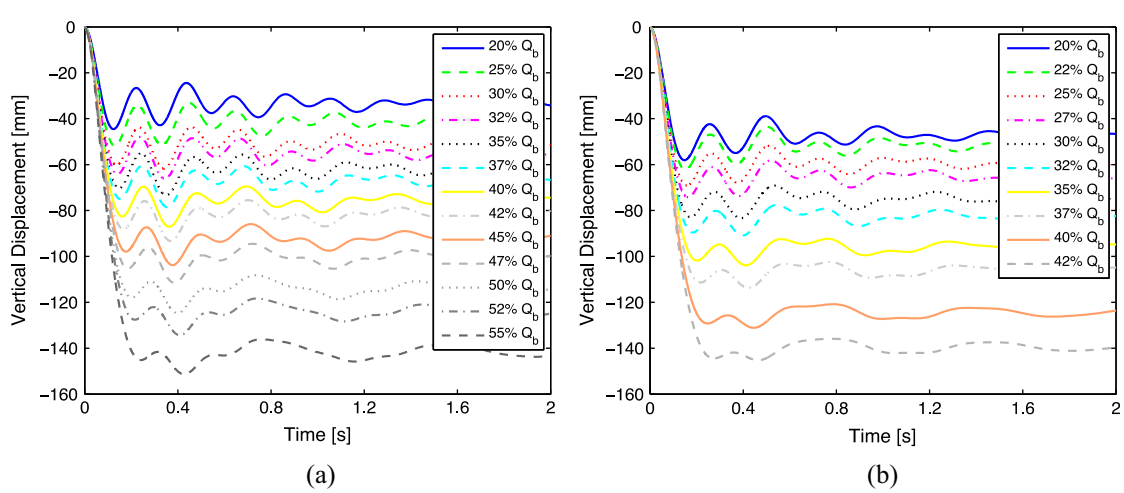

Fig. 7. Vertical displacement time-histories of 2D models: (a) EC8-conforming building; (b) EC2-conforming building.

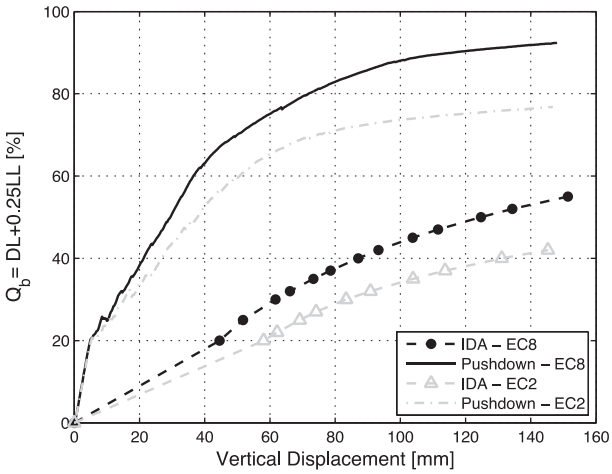

Fig. 8. IDA and pushdown capacity curves of 2D models.

mechanical measure that allows one to quantify the inaccuracy of pushdown analysis for different design targets beyond the elastic range of the structure.

As one of the main objectives of this preliminary study was to have an estimate of this quantity, pushdown analysis was carried out by applying uniform downward loads to the entire structure in accordance with GSA (General Services Administration) prescriptions [13]. This load distribution is indeed consistent with that assumed for vertical mass when performing dynamic simulations. Nevertheless other contributions (see e.g. [10,13]) suggested more complex and accurate techniques to conduct pushdown analysis. In particular, the Unified Facility Criteria (UFC) released in 2009 [14] proposed the application of non-uniform load distribution to the structure, identifying adjacent spans and areas away from the removed column. Therefore this approach, which is aimed to equivalently include dynamic effects in a static fashion whether nonlinear static procedures are adopted, was tried in comparison with IDA and pushdown according to GSA provisions. As shown in Fig. 12, where the capacity curves of the EC8-compliant 2D model are collected, UFC is more conservative than GSA, but it still overestimates the progressive collapse potential of the structure. In light of these considerations, the dynamic nature of progressive collapse phenomena was explicitly included in the probabilistic simulations to develop the fragility models proposed in the following.

4. Fragility analysis

In the second stage of this research, the numerical methods for progressive collapse resistance assessment were implemented in a fragility analysis framework to provide the probability of exceedance of different damage states induced by column loss events. Even though this proposal may be considered quite a novel method for its potential applications and implications, the approach shown and discussed in the following is based on well-established tools that were developed and validated either in progressive collapse or earthquake engineering: fiber-based models (see e.g. [18,31]), incremental dynamic analyses (e.g. [44,45]) and methodology of fragility analysis (e.g. [35–39]).

First of all, some design parameters were assumed to be RVs so their uncertainty was characterized by means of probability distribution functions for each building class under study. Then, performance criteria associated with different damage states were distinctly defined for each building class. Finally, fragility curves were developed to provide the probability of exceeding a given limit state in terms of vertical beam drift (global level) or concrete/steel strain (local level) using IDA. As shown in Fig. 13, the key component of the methodology involves the definition of the capacity for each structure realization which was randomly generated within the EC2-conforming or EC8-conforming building class. Capacity was assessed through incremental NLTHA to obtain a multiplier of the code-compliant downward load Q_b for progressive collapse application (see Eq. (2) [13]), which is herein assumed as IM. The definition of the scale factors (i.e. points on the IDA capacity curve) corresponding to different limit states is described in Section 4.2. Statistics and probability distribution functions for design loads, geometric characteristics and material properties, which were assumed in Monte Carlo simulation to generate 2D and 3D models representative of each building class, are discussed in Section 4.1.

4.1. Case-study building classes and uncertainty modeling

Consistently to the comparative analysis approach discussed in Section 3, fragility analysis was carried out on two populations of RC framed structures, as follows:

- EC8-conforming building class (i.e. earthquake resistant buildings);
- EC2-conforming building class (i.e. gravity-load designed buildings).

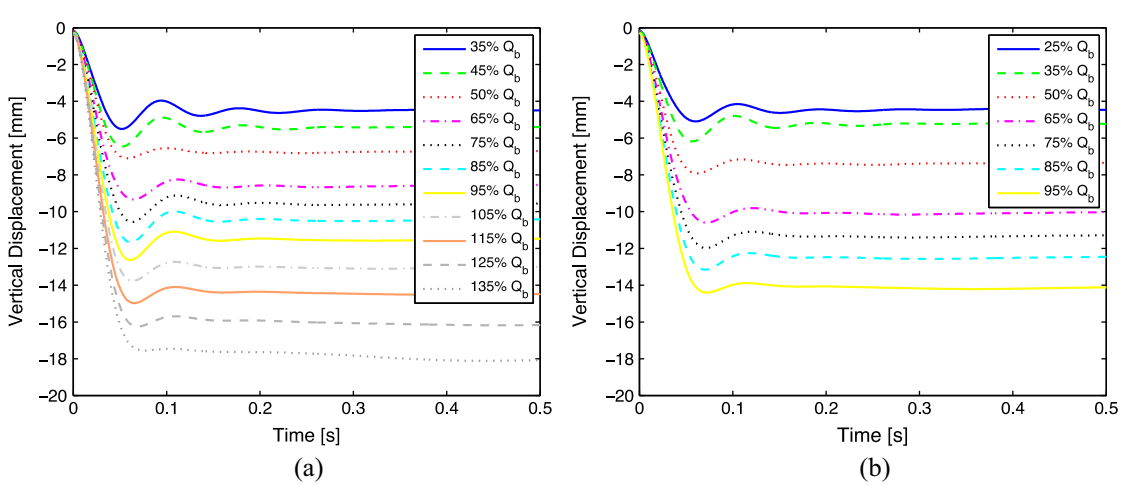

Fig. 9. Vertical displacement time-histories of 3D models: (a) EC8-conforming building; (b) EC2-conforming building.

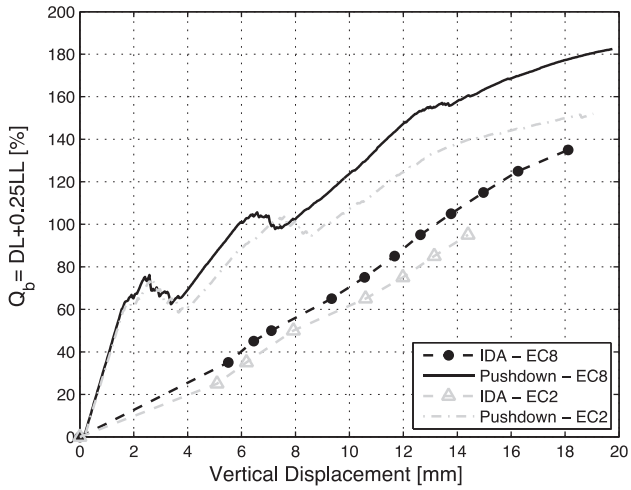

Fig. 10. IDA and pushdown capacity curves of 3D models.

Two types of structural models of were randomly generated and analyzed in a probabilistic fashion, namely 2D and 3D models for each building class under study. As a result, a total amount of four fragility models corresponding to as many combinations in terms of code regulations for design and structural modeling strategies were developed.

First of all, uncertainties in geometry, material properties and loads were modeled. According to past studies [35,36,38,39], the span lengths in the longitudinal and transverse directions, namely L_x and L_y, were assumed as RVs with uniform distribution in the range [4.0 m, 6.0 m]. By contrast, the interstory height was considered a deterministic parameter equal to 3.0 m, as progressive collapse resistance is almost insensitive to it. Material uncertainties were taken into account by randomly selecting steel yielding strengths of 380 MPa and 450 MPa and cubic concrete strengths of 25 MPa, 30 MPa and 35 MPa, each of them assumed to have the same probability of occurrence [38]. Those values were considered as nominal strengths which were multiplied by a normally distributed (dimensionless) RV with mean equal to 1.1 in case of reinforcing steel and 1.5 in case of concrete, and coefficient of variation (CoV) equal to 10%. Finally, a normal distribution was assigned to dead load with mean of 3.0 kN/m^2 and CoV = 17%, whereas live load was considered as deterministic parameter equal to 2.0 kN/m^2 because it does not have significant influence on the overall downward load for progressive collapse applications, as expressed by Eq. (2). In this first application, both wind and snow loads were considered as deterministic parameters with values equal to those adopted for the deterministic prototypes discussed in Section 3. A similar consideration can be drawn for what concerns the seismic input assumed for design of the EC8-conforming building class. As suggested by Borzi et al. [38], the Latin Hypercube algorithm was used for Monte Carlo sampling of RVs with normal distribution, thus allowing a better representation of the sample in the low probability region of the distribution.

The design process was carried out considering cross section dimensions of beams and columns to be constant and equal to those assumed for the two deterministic case-study structures (see Table 1), while the amount of steel reinforcement provided to detail them was derived from a direct application of EC8 [34] and EC2 [33] provisions. Table 2 summarizes the range of variation in the reinforcement ratio of EC8-compliant and EC2-compliant building populations.

4.2. Performance criteria and damage states definition

To convolve capacity and demand under progressive collapse condition, the following three criteria have been introduced – either at local/sectional (i.e. concrete and steel strains) or global/structural (i.e. vertical drifts) levels – for limit states definition: slight damage (LS1), significant damage (LS2) and complete damage (LS3). The light damage limit condition (LS1) refers to a situation in which the building can be immediately used after an event with the need for minor repair and/or strengthening only, so it can be regarded as a sort of serviceability limit state. On the other hand, the complete damage condition (LS3) can be considered as an ultimate limit state beyond which the structure is close to collapse, being no longer able to deform and sustain any further load increment nor the gravity loads for which it has been designed. Finally, the significant damage state (LS2) can be regarded as an intermediate case beyond which the building becomes unsafe for its occupants, being the majority of its source of nonlinearity yet fully in use. Hence, it may be identified as a sort of life safety limit state, in light of conventional performance-based engineering principles [56–58], and it may be assumed as the condition beyond which the structure should be evacuated after the event. As a result of this application, it was decided to define this limit state using a global (vertical drift) rather than a local (axial strain) parameter. Nonetheless, the drift ratios specified for this limit state may be associated with conventional steel and concrete strains, as shown in [35,36,38,39].

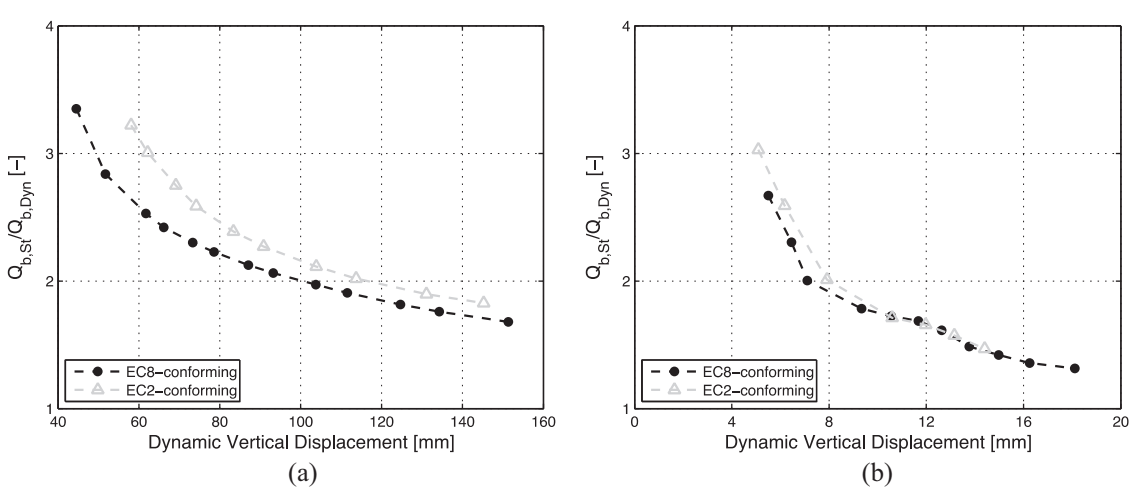

Fig. 11. DAF versus vertical displacement: (a) 2D model; (b) 3D model.

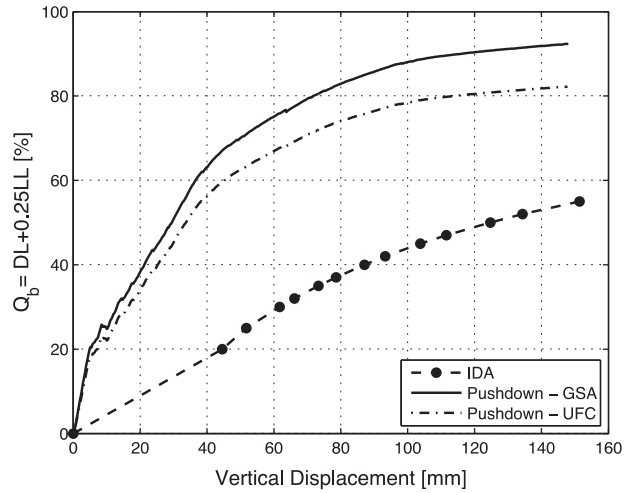

Fig. 12. Pushdown analysis: GSA [13] versus UFC [14] approach. Note: EC8-compliant 2D model.

To determine whether a frame member reached a limit condition, axial strains and vertical drifts experienced in the critical portion of the structure where the progressive collapse mechanism was forced to occur were compared to limit capacities conventionally identified as follows:

- LS1 was defined by steel and concrete strains for both EC8-conforming and EC2-conforming building classes. For each randomly generated structure, the yielding strain of steel bars (ε_{sy}) was computed as the ratio between yielding strength and Young's modulus, while concrete strain at peak strength (ε_{cp}) was determined in accordance with Mander et al. [53].
- LS2 was supposed to occur when the vertical drift (D_v), obtained as the ratio between the peak vertical displacement above the column removed and the beam span length, exceeded a deterministic threshold. In detail, that drift was assumed to be 0.5% and 1.0% in the case of EC2-conforming and EC8-conforming building classes, respectively.
- LS3 was characterized in terms of ultimate steel strain (ε_{su}) and ultimate concrete strain (ε_{cu}). For each randomly generated structure, the ultimate concrete strain was calculated according to different material properties and reinforcement arrangement [53], whereas the buckling/fracture strain of steel bars was set to 4% and 6% in the case of EC2-conforming and EC8-conforming building classes, respectively.

Table 3 outlines the limit states and threshold values of vertical drift, concrete strain, and steel strain for each building class of interest for this study. It can be seen that the use of past statistical studies [35,36,38,39] and concrete constitutive model by Mander et al. [53] led to single values or ranges for performance thresholds. Strain levels in close agreement with those shown in [36,39] or used in [35,38] to define limit states in terms of chord rotation capacity for RC members were assumed.

4.3. Derivation of fragility curves

As shown in Fig. 14, a series of IDAs were performed to characterize the nonlinear dynamic behavior of the randomly generated structures of each building class, using the same numerical techniques described in Section 3.2 in terms of geometric and material nonlinearities, element formulation and discretization, column-removal routine and strategy, and analysis algorithms. Therefore, a multiplier of Q_b was identified on each IDA curve for each damage state, in order to assemble a damage probability matrix. The latter contained the fractions of sampled structures in each damage state, for a set of increasing Q_b levels. The cumulative fraction of buildings in each damage state was then

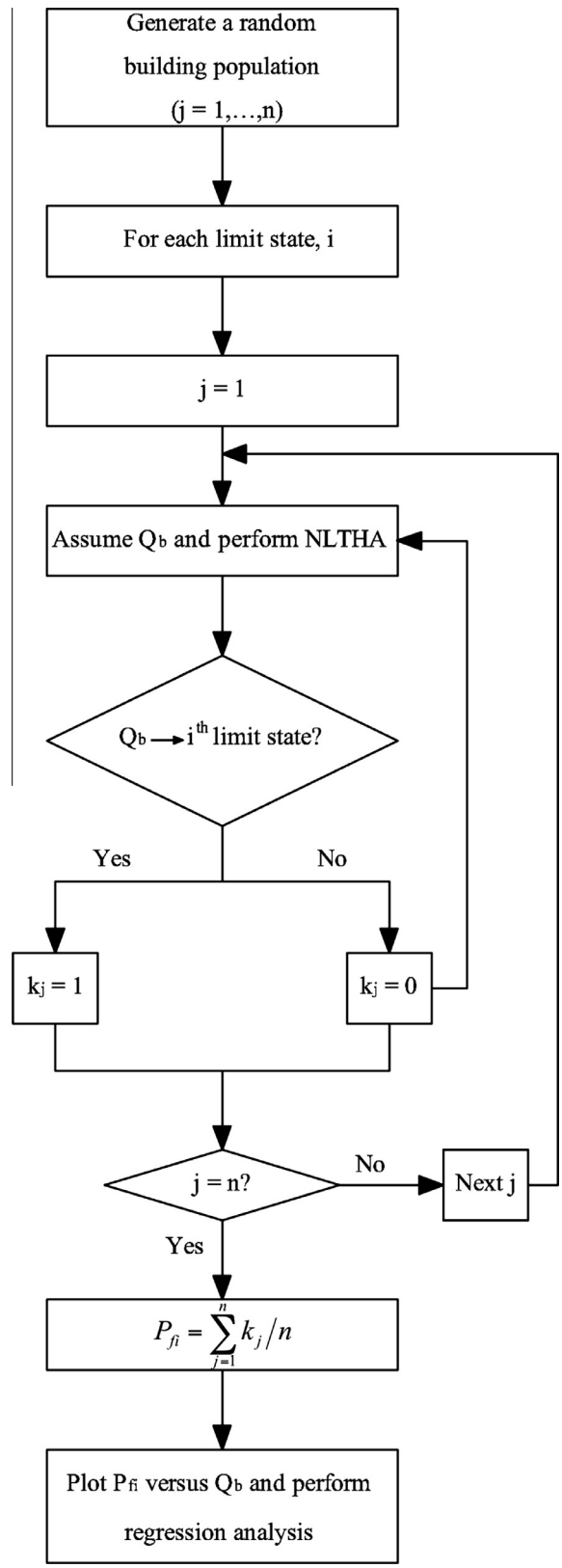

Fig. 13. Routine for derivation of fragility curves.

Table 2
Range of variation in the longitudinal reinforcement ratio of EC8-compliant and EC2-compliant building populations.

Building class	Element	Threshold	Layout	Ratio (%)
EC8-conforming	Beam	Upper bound	3Ø22 + 4Ø10 + 3Ø22	1.73
		Lower bound	3Ø18 + 4Ø10 + 3Ø18	1.23
	Column	Upper bound	12Ø22	2.85
		Lower bound	12Ø18	1.91
EC2-conforming	Beam	Upper bound	5Ø16 + 4Ø10 + 3Ø16	1.28
		Lower bound	4Ø12 + 4Ø10 + 3Ø12	0.74
	Column	Upper bound	8Ø18	1.27
		Lower bound	12Ø10	0.59

Table 3
Limit states and thresholds of EC8-conforming and EC2-conforming building classes.

Building class	Limit state	Parameter	Threshold
EC8-conforming	LS1	ε_{sy}	0.17%–0.27%
		ε_{cp}	0.21%–0.23%
	LS2	D_v	1.00%
	LS3	ε_{su}	6%
		ε_{cu}	0.7%–1.1%
EC2-conforming	LS1	ε_{sy}	0.17%–0.27%
		ε_{cp}	0.20%–0.21%
	LS2	D_v	0.5%
	LS3	ε_{su}	4%
		ε_{cu}	0.5%–0.7%

computed, summing up the percentages of frames pertaining to each of them according to Monte Carlo simulation. Finally, a lognormal cumulative distribution function was fitted to fragility points through nonlinear regression analysis, in order to provide the probability of exceeding each damage state in a continuous fashion.

Fragility functions for each building typology are shown in Figs. 15 and 16, while their parameters, in terms of mean, standard deviation and coefficient of determination, are collected in Table 4. A good match with fragility points from NLTHA can be observed for each limit state (i.e. LS1, LS2 and LS3), design approach (i.e. EC8-conforming and EC2-conforming structures) and structural idealization (i.e. 2D and 3D models), being R^2 in the range 0.839–0.999. In particular, the scatter is very low in case of

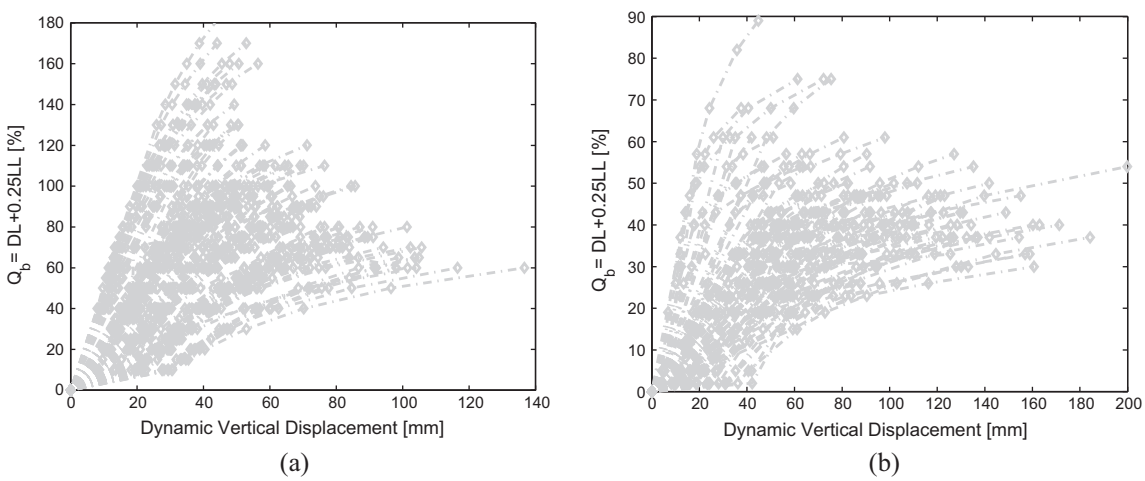

Fig. 14. IDA curves of 2D models: (a) EC8-conforming building class; (b) EC2-conforming building class.

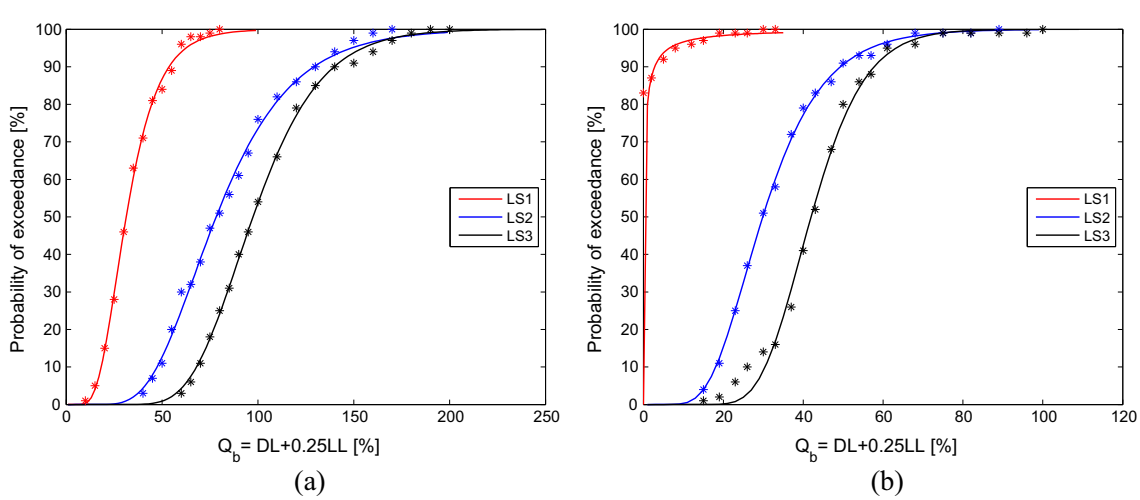

Fig. 15. Fragility curves of 2D models: (a) EC8-conforming building class; (b) EC2-conforming building class.

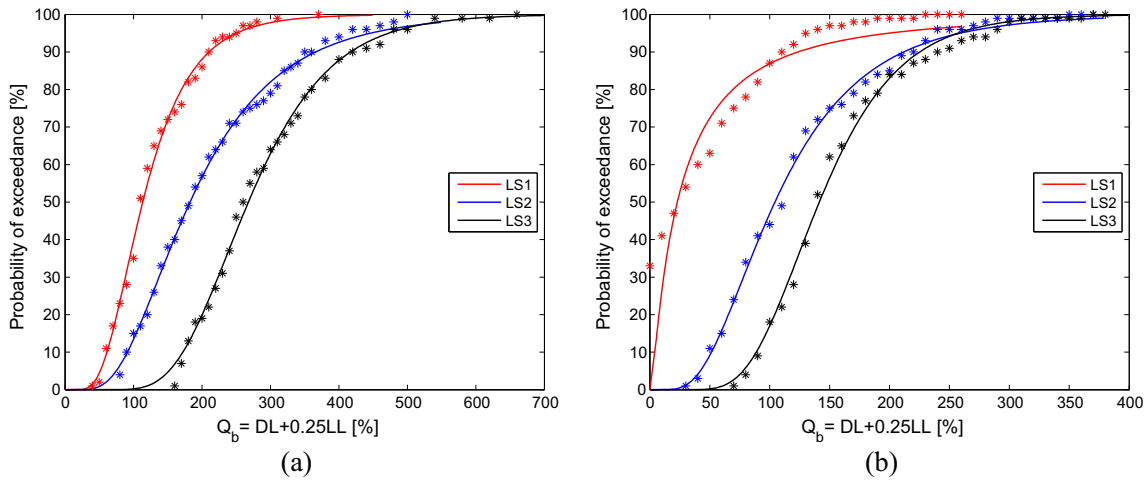

Fig. 16. Fragility curves of 3D models: (a) EC8-conforming building class; (b) EC2-conforming building class.

Table 4
Mean (μ), standard deviation (σ) and coefficient of determination (R^2) of fragility functions in terms of Q_b.

Building class	Model	LS1			LS2			LS3		
		μ	σ	R^2	μ	σ	R^2	μ	σ	R^2
EC8-conforming	2D	−1.162	0.420	0.998	−0.246	0.391	0.996	−0.026	0.281	0.999
	3D	0.126	0.482	0.995	0.601	0.543	0.997	0.985	0.345	0.996
EC2-conforming	2D	−6.548	2.323	0.955	−1.208	0.391	0.999	−0.868	0.249	0.995
	3D	−1.473	1.314	0.839	0.043	0.552	0.996	0.347	0.3631	0.995

EC8-conforming buildings, as their fragility models are characterized by a coefficient of determination close to unity (i.e. $0.995 \leqslant R^2 \leqslant 0.999$) for both 2D and 3D models. As far as the EC2-conforming building class is concerned, an identical accuracy is observed for LS2 and LS3, considering both 2D and 3D model, while the goodness of fit for LS1 was slightly lower as R^2 equals 0.955 and 0.839 for 2D and 3D models, respectively. Nevertheless, that limit state (i.e. slight damage condition) is less crucial than others for progressive collapse applications, in which (i) moderate damage is likely to occur as a consequence of redistribution of vertical loads from a removed column and (ii) an optimum design target is life safety or collapse prevention limit state in light of performance-based design principles. Moreover, the fragility models proposed herein reaffirm the key role played by secondary framing beams and seismic design/detailing criteria for a cost-effective progressive collapse vulnerability mitigation of RC structures, as discussed in Section 3. In fact, a symmetrical reinforcement configuration is effective for the creation of a rationally-controlled primary resisting mechanism, whereas redundancy added by secondary beam systems (designed to comply with minimum reinforcement requirements prescribed in current codes) is crucial to ensure alternative load paths under extreme loading conditions due to sudden column losses. In this regard, it is emphasized that additional robustness resources provided by secondary frame systems may be predicted in the design stage, for instance through a correlation between the maximum vertical displacement of the 3D model normalized to that of the 2D model ($D_{vmax,3D}/D_{vmax,2D}$) and the maximum demand-to-capacity ratio of beams [18].

5. Conclusions

In this study, a procedure for large displacement inelastic dynamic analysis of RC framed structures subjected to threat-independent sudden column loss was integrated in a probabilistic framework for the derivation of fragility functions. Two low-rise RC building classes were investigated, namely gravity-load-resistant RC buildings designed in compliance with EC2 and earthquake-resistant RC buildings designed in compliance with EC8. A special purpose death element routine was implemented into an open FE platform to operate member removal. The case-study structures were analyzed by means of two analysis methods, i.e. pushdown analysis and incremental-mass nonlinear time-history analysis, and two modeling strategies, i.e. 2D and 3D fiber force-based models. Nonlinear analyses allowed the prediction of the overall structural response, resisting mechanisms and damage patterns under column loss scenarios.

First of all, the authors assessed the implications of different design rules, modeling strategies and nonlinear analysis techniques on progressive collapse assessment for two 4-story buildings with deterministic features and the same overall geometry, which were separately designed in accordance with EC2 and EC8. Comparing response predictions for those structures have highlighted that: (i) earthquake-resistant design increased robustness of approximately 20–40%, (ii) incremental dynamic analysis was more conservative than pushdown analysis for progressive collapse assessment, and (iii) secondary beams (included in 3D models) significantly increased the robustness of the case-study structures as they effectively contribute in controlling the overall structural response and provide an alternative load path for column loss-induced overloads.

In light of those preliminary results, the IDA approach was selected and used for fragility analysis in order to assess the probability of exceedance of different damage states given that a column loss event is occurred. The progressive collapse fragility was evaluated for both the case-study building classes, considering 2D and 3D models as well as uncertainties related to geometry of structural members, material properties and gravity loads. Classical performance-based assessment principles were extended

to define demand and capacity at both local and global structural levels. A number of fragility models were then derived, allowing the following conclusions to be drawn:

- The fragility models proposed in this study are optimally fitted to discrete fragility estimates provided by IDA combined with Monte Carlo simulation, particularly for life safety and total collapse prevention limit states which are crucial for progressive collapse assessment.
- Different fragility levels are associated with multiple damage states, depending on the building class and modeling strategy used for probabilistic progressive collapse assessment.
- Significant benefits from both seismic design and secondary beams on the actual robustness level of RC buildings are confirmed and quantified by the fragility functions proposed herein.

The fragility analysis methodology applied to low-rise RC building structures may be extended to other RC building classes and structural types, explicitly incorporating the contribution from further secondary systems such as slabs and infill walls. The probabilistic approach integrated with advanced computational strategies is an effective tool for rational and quantitative risk assessment of structures that are prone to progressive collapse under natural or man-made extreme events. Further research may focus on fragility of strategic buildings designed for progressive collapse in accordance with specific guidelines and codes.

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